

# TrES-5b

R-0987

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_180713 · created 2026-07-29T22:24:44+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458312.8788 +/- 0.0026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.179 +/- 0.055                 |
| Transit depth (Rp/Rs)^2                           | 3.2 +/- 1.95 [%]                |
| Orbital Inclination (inc)                         | 80.73 +/- 2.89                  |
| Airmass coefficient 1 (a1)                        | 32.62 +/- 17.75                 |
| Airmass coefficient 2 (a2)                        | -3.0 +/- 0.84                   |
| Scatter in the residuals of the lightcurve fit is | 49.86 %                         |
| Best Comparison Star                              | #2 - [164, 264]                 |
| Optimal Aperture                                  | 2.05                            |
| Optimal Annulus                                   | 6.5                             |
| Transit Duration (day)                            | 0.042 +/- 0.03                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.179 $\pm$ 0.055 vs 0.142 (deviation 0.61 $\sigma$ ) |
| Duration fit vs published            | 0.042 d vs 0.0758 d (deviation 1.03 $\sigma$ )        |
| Mid-time O-C                         | -5.6 min                                              |
| Depth SNR                            | 3.0                                                   |
| Residual scatter                     | 49.86 %                                               |

### Light curve

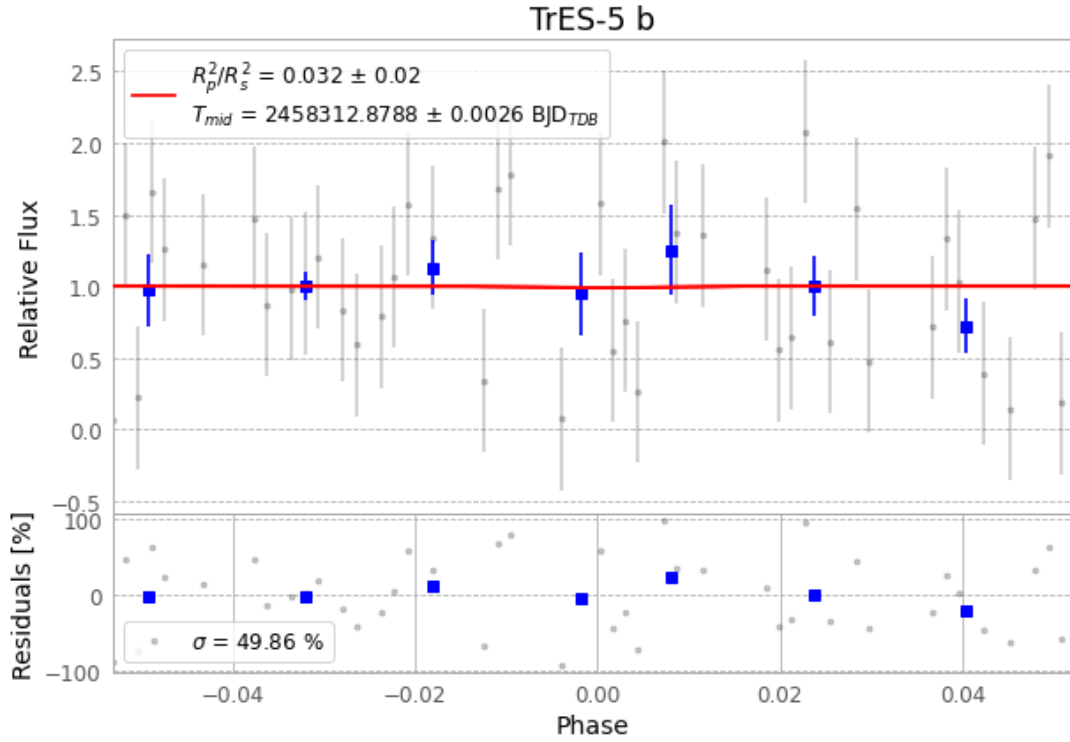


Figure 1 — FinalLightCurve\_TrES-5 b\_2018-07-13.png (SHA-256 ad89c36b59dc9a0d...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [181, 221], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d14c96df3701c8d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

78 FITS frames (51 MB), TRES-5180713070912.FITS → TRES-5180713110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-180713.log (b333abd3a814...), FinalLightCurve\_TrES-5 b\_2018-07-13.png (ad89c36b59dc...), FinalLightCurve\_TrES-5 b\_2018-07-13.csv (068d4ba73fa9...)

## Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 22:24Z.